

## Perceptions of urban stream corridors within the greenway system of Sapporo, Japan

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### Abstract

Many rivers and streams have disappeared through flood control, drainage works and urban development in Sapporo, Japan. Also, accessibility to and naturalness of the streams have deteriorated. Newer river and stream corridor improvement technologies such as nature-oriented construction, and public involvement methods are emphasized in the Sapporo General Green Space Plan. The purpose of this study is to make clear the perceptions of stream corridors and to show some factors which are important for integrating them into the Urban Greenway Plan.

After consideration of landforms, stream corridor land uses and location, five stream corridors which have been improved in the Sapporo urban areas were selected. Neighborhood residents' perceptions of the stream corridors were surveyed using a questionnaire, and a census of the vegetation along the stream banks was also taken.

The results showed that improvement of the stream corridors in question had been supported by nearby residents. After the factor analysis, five important factors of stream corridor perception, namely, "recreational use," "participation," "nature and scenery," "sanitary maintenance," and "water safety" were identified. Based on these factors, stream corridor features including vegetation that may affect preference were discussed as variables for integrating the stream corridors into the Greenway Plan. Furthermore, using the factor scores, respondents were classified into three groups: the Passive Group, the Recreation Interested Group, and the Participation Oriented Group. Corresponding to these groups, a number of programs facilitating participation in or support of greenway planning are suggested.

In conclusion, it is suggested that (1) enhancing the ecological environment; (2) creating a recreational circulation system; and (3) making full use of the natural and cultural characteristics of the local areas are important to the integration of stream corridors in urban greenways.

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### 1. Introduction

Nearly all major cities have been built along river corridors (Baschak and Brown, 1994). Recently, river corridors are being focused on as important natural networks as well as cultural and recreational resources

(e.g. Baschak and Brown, 1994, 1995; Cook, 1991). In Japan many large cities are located on alluvium plains that were formed by, and covered with many rivers and smaller streams. However, many of these have disappeared or deteriorated through flood control, drainage works and urban development. Japan frequently experiences heavy rainfalls, which in turn causes flooding. Many streams have thusly been channelized, and flood control measures have destroyed the stream corridors'

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natural environment and public accessibility to the water. High water channel terraces have often been used for sports and other recreational pursuits due to a lack of open spaces in densely developed urban areas, which in themselves account for an additional factor in the deterioration of the natural environments of rivers. Until now, a number of greenways in cities including river corridors have been planned in Japan, however, it was not easy to realize these plans because of the original provisions of the Japanese River Law which set out strict conditions for flood control, and creates river improvements without considering the natural environment or residents' demands.

Sapporo, one of the largest cities in Japan, with a population of about 1.8 million, was built mainly on the alluvium fan of the Toyohira River. Developed areas have expanded across northern and eastern low lands and southern and western hillsides. The development of farmland and continued urbanization deteriorated many streams, and many springs, which provided groundwater for those streams, dried up. In spite of this, however, there are about 300 rivers, streams, and subsoil waters including the Toyohira River and its tributaries which flow through the city (City of Sapporo, 1982). While a large flood in 1981 showed that the northern and the eastern lowlands have a high potential for flooding, the revision of the Japanese River Law in 1997 directed attention to the importance of rivers' environmental value in addition to flood control. The revised act requires citizen participation in the planning process and management, promoting naturalized restoration of river corridors. As well, some of the practices used in river corridor improvement have been changing radically.

Recently, improved river and stream corridor techniques such as nature-oriented construction and public involvement methods have been emphasized in the Sapporo General Green Space Plan (City of Sapporo, 1999). Within the General Green Space Plan, the Greenway Plan section intended that parks and green areas should be systematically located and connected by bicycle paths and trails on riverbanks and green zones. Many of the greenways are based on river corridors (Fig. 1). The proposed plan seeks to improve these riverside greenways and upgrade the Toyohira River so that it becomes a more attractive river. Smaller rivers will also receive similar attention for the purpose of serving as a riverside greenway

linking various sized parks and green areas in the urbanized area. However in the city-wide plan, small stream corridors are not described well because of their difference in scale; it is thought that the evaluation of the deteriorated stream corridors is not a high enough priority, and some were not perceived well by residents (Lee and Asakawa, 1991).

To inform this study, a review of perception studies of urban stream corridors was conducted. Harashina et al. (1989) focused on perceptions of flood control measures, and showed that importance of residents' knowledge regarding water quality and flood danger in respect to river improvement. Kuroyanagi et al. (1993) showed that trails for strolling, spaces for relaxation, and good scenery were called for by residents living near the rivers. Hashimoto and Ejiri (1993) classified residents' attitudes toward rivers from the points of view of "dwelling with respect to floods, etc.," "visual," and "ecological." House and Fordham (1997) identified public perceptions of river corridors and attitudes toward river works. Tapsell (1995) indicated that river restoration was substantially supported by residents. Regarding the effect of river landscape, Kaplan (1977, p. 285) comments that "even unspectacular rivers provide a source of enjoyment and tranquility for many who use only the riverbanks, view the river from afar, or who only know that it is there and available."

Kaplan (1977) showed that rivers and riversides by their nature were appreciated by people when they provide both a sense of orderliness and a sense of involvement and mystery. Furthermore, Lee (1980) applied Kaplan's (1975) information theory to the components of land, vegetation, and water for the Louisiana River landscape. Although urban streams were not included, Herzog (1985) indicated that the predictor variables in determining preference differ by types of waterscape. Although, many studies showed that natural landscapes with plants are preferred by people (e.g. Kaplan and Kaplan, 1989; Ulrich, 1986; Sardon, 1988), the preference may depend on the types of vegetation. Some preferred vegetation types such as widely spaced trees with an even ground-cover, are outlined by the information theory (Kaplan and Kaplan, 1989) and the evolutionary theory (Ulrich, 1986). Meanwhile as Nassauer (1997) showed, cultural evaluation such as controlled or weed-less vegetation may be preferred. When stream corridors are restored, formulating a planting plan and the mainte-

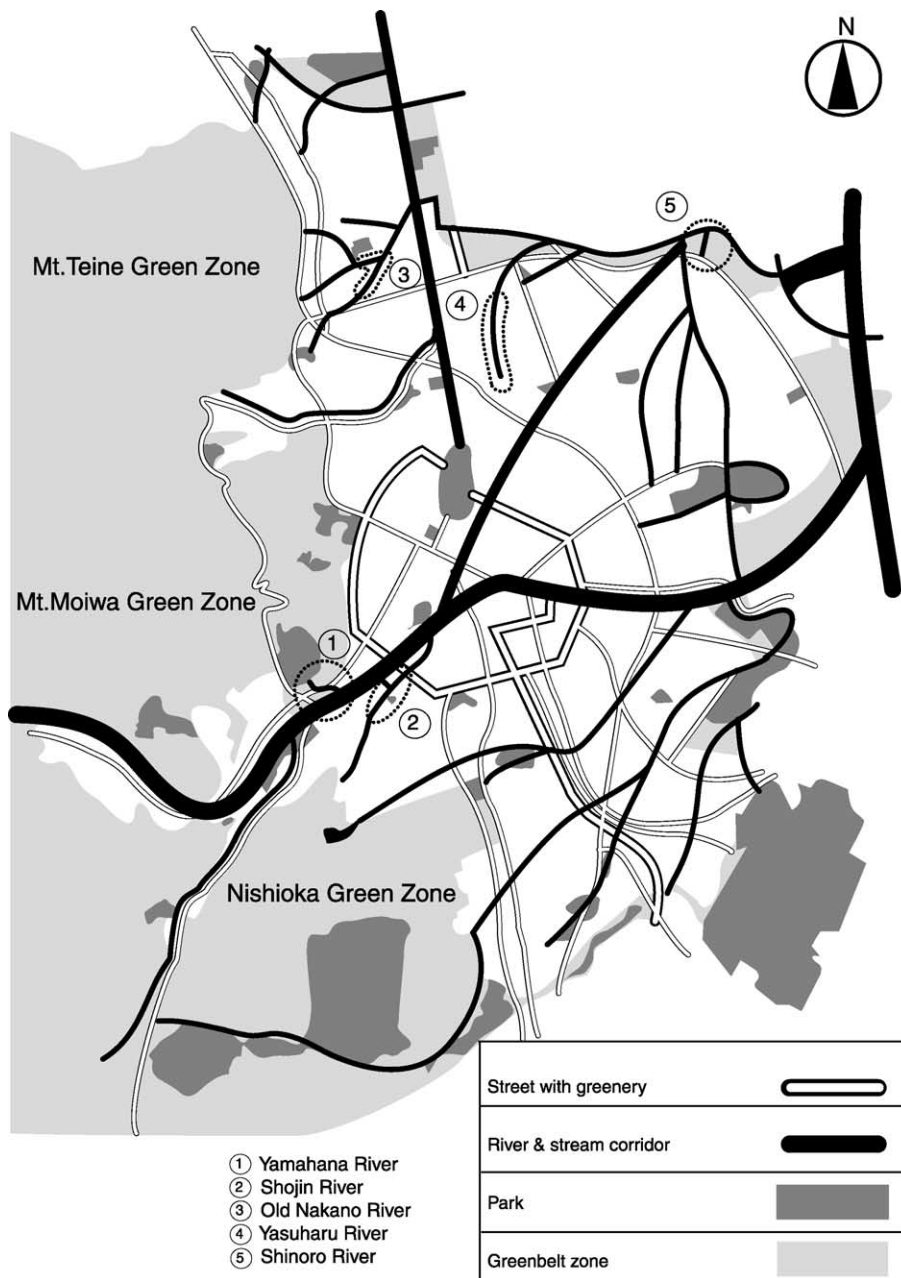


Fig. 1. The Sapporo Greenway Plan and the locations of the surveyed stream corridors.

nance of vegetation are important and difficult tasks due to the unclear relationship between naturalness and peoples' preferences. As well, change in the natural vegetation is not so easy to control.

Citizen participation is an integral piece of nearly every project or process that environmental design professionals undertake (Jones, 1999). Furthermore, citizen participation in the management of urban

stream corridors is being seen as an efficient tool by the Sapporo municipal administration (City of Sapporo, 1999).

In regard to greenway planning, Shannon et al. (1995) showed the process used to identify and assess visual and scenic resources in the St. Laurence River Valley as a foundation for greenway planning. However, few studies concerning residents' perceptions of urban stream corridors for greenway planning have been conducted.

This study focuses on assessing the restorations of stream corridors in the urban Sapporo area which were improved or restored before the revision of the River Law. The purpose of this study, in turn, was to obtain information on residents' perceptions of these urban stream corridors. In particular, the study looked at factors including vegetation which influence preference, as well as attitudes toward participation in stream corridor improvement. After investigating the effects of stream improvements on the residents' perceptions, a basic discussion for integration of the stream corridors into the Greenway Plan is offered.

## 2. Methods

Five stream corridors that were restored for ecological or recreational accessibility in the urban Sapporo area were selected for this study (Table 1). Fig. 1 shows the Sapporo Greenway Plan and the locations of the five stream corridors, which are surveyed in this study. Two of them are located on the upper or the middle section of the alluvial fan and the other three are located on the northern lowlands. Fig. 2 shows the structures of these stream corridors. Five nearby residential areas within 500 m of the streams were selected and a

questionnaire was delivered to 675 randomly chosen households situated in those areas in May 2001. The number of questionnaires distributed was decided on based on the population densities of the survey areas. The completed questionnaire was collected from 415 residents. Survey respondent numbers for each area are shown in Table 2. In this survey, residents were asked about their perceptions of their nearby stream corridor by rating 25 items on a five-point Likert scale. A response value of "5" represented the highest or most positive response while a "1" represented the most negative assessment or lowest response. Respondents were also asked about how frequently they used the streams in addition to questions about their maintenance and residents' background such as their age, gender, and length of residence in the area.

To obtain information on the factor structure of residents' perceptions of urban streams and their corridors, a factor analysis using a correlation coefficients matrix of the 25 perception items was conducted. The principle axis method, based on the criterion of selecting eigenvalues greater than one, and varimax rotation of the factors, was used. An analysis of variance (ANOVA) using standardized respondent factor scores (mean = 0, and S.D. = 1) was conducted to show the mean differences of the factor scores in the five survey areas (the stream corridors). Using a Euclid distance matrix of standardized respondent factor scores (mean = 0, and S.D. = 1), Ward's method of cluster analysis was conducted to divide the respondents into groups based on the factor structure of perception.

In addition to the questionnaire, the vegetation surrounding the streams, namely, the low water channel terrace, the high water channel terrace and the bank slope areas was also surveyed. As shown in Fig. 3, along each of the five streams, four survey sites were selected from the upstream to downstream areas, in which about five (1 m<sup>2</sup>) quadrats were placed, respectively. All species, which occurred in the quadrats, were listed and the coverage (%) of each species was measured in each lot. In addition to the average population numbers for all species combined, this study also used the respective average population numbers for native and exotic species as well as the ratio of exotic species to all species as indices of biodiversity or naturalness and disturbance of vegetation.

Table 1  
Improvement period and coverage of the surveyed stream corridors

| Survey area      | Improvement period | Improvement coverage (km) | Total length (km) |
|------------------|--------------------|---------------------------|-------------------|
| Yamahana River   | 1990–1994          | 1.5                       | 3.0               |
| Shojin River     | 1971–1976          | 3.5                       | 14.2              |
| Yasuharu River   | 1990–1994          | 0.8                       | 5.5               |
| Old Nakano River | 1993–1996          | 1.0                       | 4.8               |
| Shinoro River    | 1984–1999          | 3.1                       | 3.1               |

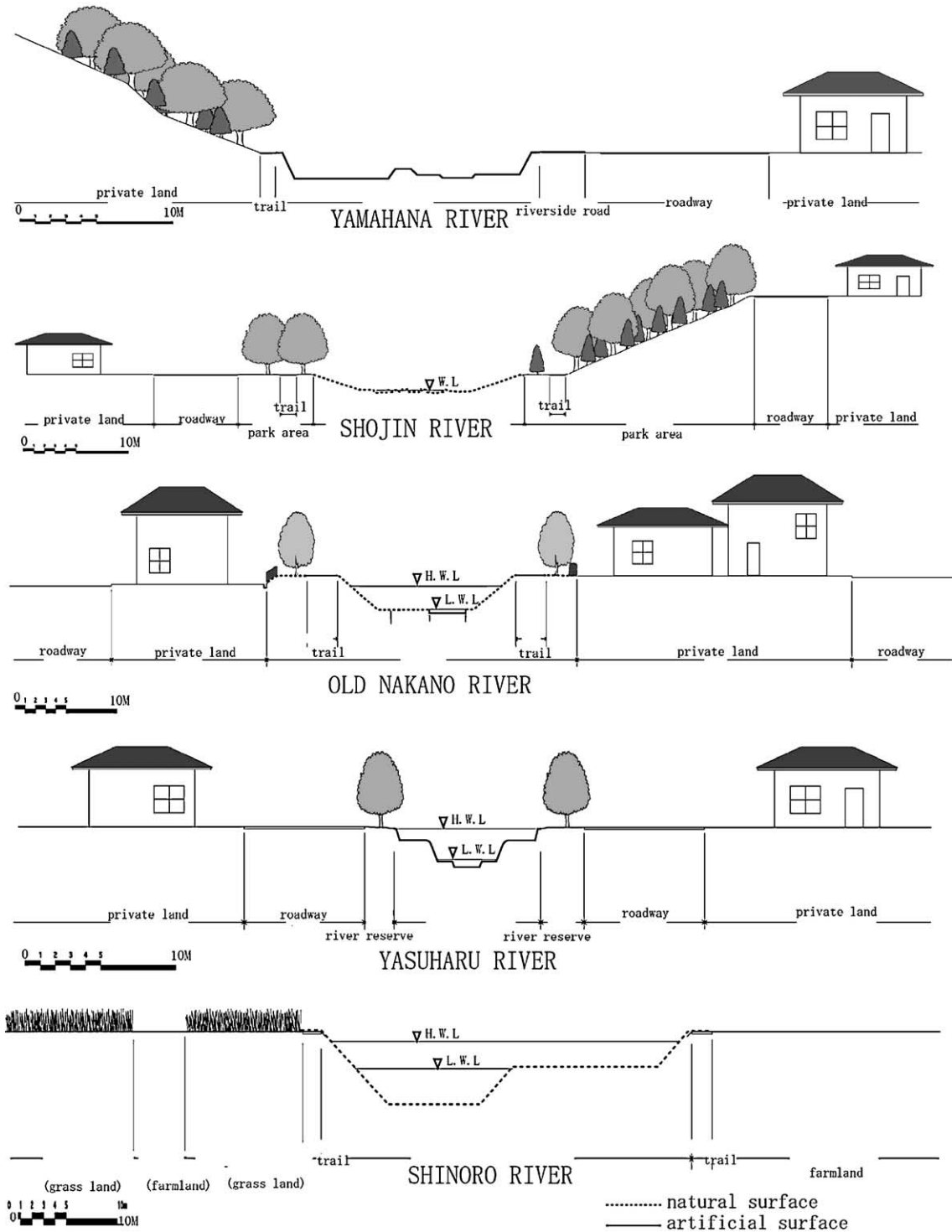


Fig. 2. Structures of the surveyed stream corridors.

Table 2

The survey areas and numbers of respondents

| Name of area     | Number of questionnaires |          | Ratio (%) | Zoning and location                                         |
|------------------|--------------------------|----------|-----------|-------------------------------------------------------------|
|                  | Distributed              | Received |           |                                                             |
| Yamahana River   | 105                      | 59       | 56        | Residential area near CBD, on the upper of the alluvial fan |
| Shojin River     | 105                      | 65       | 62        | Residential area near CBD, on the mid-alluvial fan          |
| Old Nakano River | 229                      | 136      | 59        | Residential area, in the north western lowland              |
| Yasuharu River   | 177                      | 102      | 58        | Residential area, in the northern lowlands                  |
| Shinoro River    | 59                       | 53       | 75        | Urban fringe, in the northern lowlands                      |
| Total            | 675                      | 415      | 62        |                                                             |

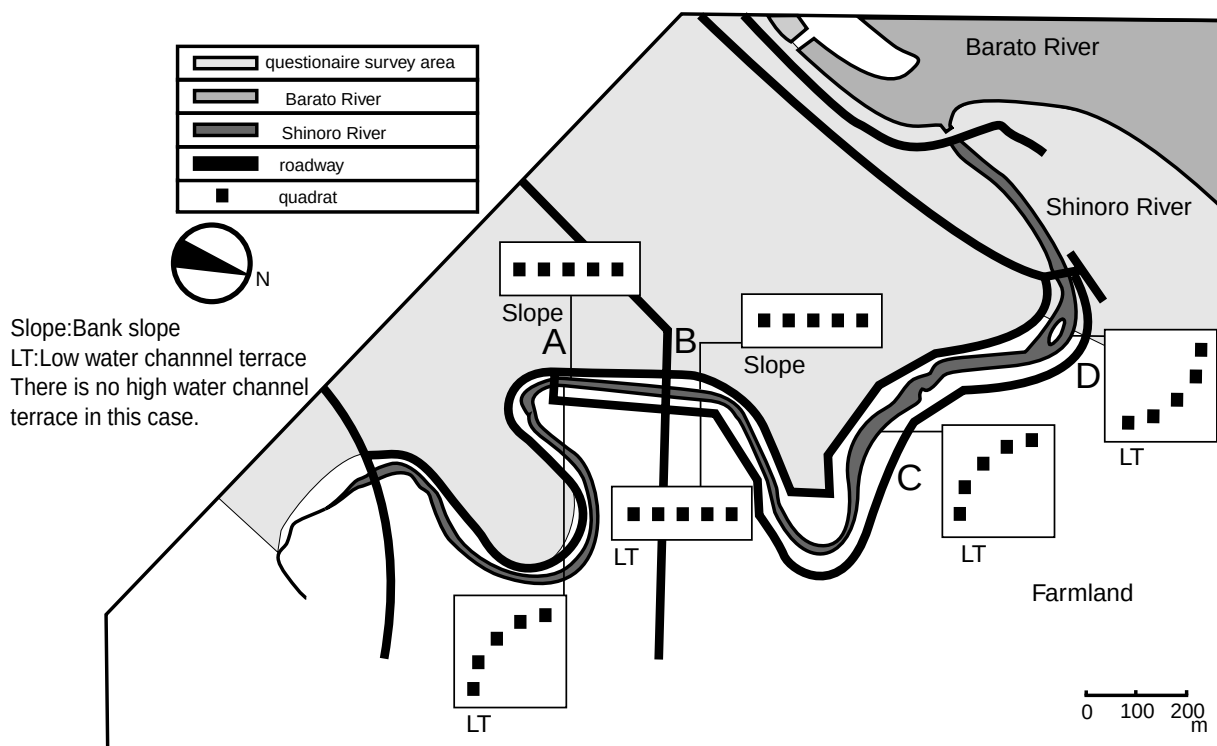


Fig. 3. An example of the setting of survey sites, vegetation survey quadrat and the distribution of questionnaires.

### 3. Results

#### 3.1. Respondents' background and perceptions results in brief

The following findings from the respondents provide a background for the analysis in this paper. Forty-two percent of the respondents were men and

54% were women. The age profiles of the respondents were as follows: 15–29 years old (9%), 30–49 years old (35%), 50–59 years old (20%), and over 59 years old (33%). In terms of length of residence in the survey area, 18% of respondents had resided there for under 5 years, 16% from 5 to 10 years, 22% from 11 to 20 years, and 39% for over 20 years. As for the dwelling types in which respondents lived, about

70% were individually owned houses and the others were apartments.

Mean ratings of the 25 items shown in Table 3 indicated that many residents felt that the nearby stream corridors were nice places for strolling and resting (mean = 4.30), and also that accessibility to the streams was good (mean = 4.30). Residents had a relatively great interest in the stream corridors (mean = 4.00) and they felt that the stream corridors were an important element of their environment (mean = 4.40). On the other hands, it was also shown that litter, pollution of the water, and safety (from crime) were their main concerns, rating 2.5, 2.60, and 2.70, respectively. The mean score of water volume of the stream was relatively low at three points. Two stream corridors, namely, the Yasuharu and the Old Nakano Rivers, had low ratings on this scale as well as low ratings on the water quality scale. Twenty-eight percent of the respondents had participated in some kind of maintenance activities and 44% of the respondents expressed desire to participate in such activities. It is worthy to note that more than half of the respondents (55%) felt that the stream corridors should have as high a biodiversity as possible.

### 3.2. Stream corridor perceptions and preference

Based on a correlation of perception item coefficients, the “preference” item (I like the stream corridor: mean = 3.90) was found to be moderately correlated with the following items: “a good place for strolling and resting (0.57),” “good accessibility (0.46),” “well used (0.50),” “well maintained (0.44),” “good natural environment (0.41),” “good scenery (0.48),” and “important element of the environment (0.43).” The “preference” item was also correlated with “good area to live (0.61).” The “water quality,” “water quantity,” and “no litter” items did not have such high correlations with “preference” (0.27, 0.23, and 0.12, respectively). These results show that recreational use, and naturalness and scenery have an influence on preference. It also reflects the importance of good maintenance in the preference item. The relatively low correlation coefficient between the “well maintained” item and the “good natural environment” item (0.29) implies that there may be some conflicts about how to best maintain stream corridors without damaging the natural environment.

Based on the factor analysis of a matrix of the 25 perception items, six factors were found that accounted for 58% of the total variance among all judgments. Factor loadings and communalities for the items are shown in Table 3. Based on an inspection of the factor loadings, the factors were labeled as follows: factor 1, accounting for 13% of the total variance, was called “recreational use;” factor 2, accounting for 12%, was called “participation;” factor 3, accounting for 12%, was called “nature and scenery;” factor 4, accounting for 10%, was called “sanitary maintenance;” factor 5, accounting for 7%, was termed “water safety;” and the sixth factor, accounting for 5%, was termed “lighting.” The meaning of factor six is not so clear as only one item had relatively high loading.

“Good scenery” and “good natural environment” had a relatively high correlation coefficient of 0.47, and both items are included in factor 3 (nature and scenery). It should be noted that the preference item had relatively high loadings on factors 1 and 3. Thus, it is clear that the recreational use, as well as nature and scenery of the stream corridors are important factors for nearby residents. Using standardized factor scores, the analysis of variance showed the four factors of recreational use, nature and scenery, sanitary maintenance, and water safety as being statistically significantly different among the survey areas.

As shown in Table 4, the mean factor score of recreational use was high at the Yasuharu River and low at the Shinoro River. The mean score of nature and scenery was high at the Shojin River and low at the Yasuharu River. Similarly, the mean score of sanitary maintenance was high at the Yamahana River and low at the Old Nakano River. The mean score of water safety was low at the Shinoro River. Variance analysis results of a number of important greenway items, such as “connectivity of greenery,” “good scenery,” and “good natural environment” are also shown in Table 4. Although “good scenery” had no significant difference among the survey areas, significant differences did appear for “good natural environment,” with the lowest mean score occurring at the Yasuharu River. The Shojin River, which has a riparian forest, had the highest score for “connectivity of greenery,” while the Old Nakano River had the lowest score on this item. The Shojin River had the highest preference score, with the Shinoro and Old Nakano Rivers scoring on the lower end on this scale.

Table 3  
Perceptions of stream corridors: factor solution

| Items                                                   | Recreational use | Participation | Nature and scenery | Sanitary management | Water safety | Lighting | Communality | Mean |
|---------------------------------------------------------|------------------|---------------|--------------------|---------------------|--------------|----------|-------------|------|
| Recreational use                                        |                  |               |                    |                     |              |          |             |      |
| 1. The s.c. is good for strolling and relaxation        | 0.75             |               |                    |                     |              |          | 0.67        | 4.30 |
| 2. The s.c. is well used                                | 0.73             |               |                    |                     |              |          | 0.64        | 3.60 |
| 3. The s.c. has good accessibility                      | 0.70             |               |                    |                     |              |          | 0.59        | 4.30 |
| Participation                                           |                  |               |                    |                     |              |          |             |      |
| 4. I like to participate in the maintenance of the s.c. |                  | 0.83          |                    |                     |              |          | 0.74        | 3.60 |
| 5. I like to participate in the planning of s.c.        |                  | 0.81          |                    |                     |              |          | 0.69        | 3.50 |
| 6. I am interested in the s.c.                          |                  | 0.72          |                    |                     |              |          | 0.59        | 4.00 |
| 7. The s.c. is an important environmental element       | 0.44             | 0.52          |                    |                     |              |          | 0.49        | 4.40 |
| 8. I like to prevent water pollution                    |                  | 0.52          |                    |                     |              |          | 0.34        | 4.50 |
| Nature and scenery                                      |                  |               |                    |                     |              |          |             |      |
| 9. The s.c.'s greenery is interconnected well           |                  |               | 0.76               |                     |              |          | 0.64        | 3.20 |
| 10. The s.c. has good scenery                           |                  |               | 0.72               |                     |              |          | 0.61        | 3.30 |
| 11. The s.c. has good natural environment               |                  |               | 0.72               |                     |              |          | 0.58        | 3.10 |
| 12. The s.c. is good place for observation of nature    |                  |               | 0.66               |                     |              |          | 0.58        | 3.10 |
| Sanitary maintenance                                    |                  |               |                    |                     |              |          |             |      |
| 13. The s.c. has little litter                          |                  |               |                    | 0.74                |              |          | 0.58        | 2.50 |
| 14. There is little vermin                              |                  |               |                    | 0.67                |              |          | 0.58        | 3.00 |
| 15. The stream has high water quality                   |                  |               |                    | 0.64                |              |          | 0.57        | 2.60 |
| 16. There is no crime                                   |                  |               |                    | 0.64                |              |          | 0.49        | 2.70 |
| 17. The s.c. is well maintained                         |                  |               |                    | 0.50                |              |          | 0.48        | 3.40 |
| Water safety                                            |                  |               |                    |                     |              |          |             |      |
| 18. There is little danger of flooding                  |                  |               |                    |                     | 0.84         |          | 0.68        | 3.30 |
| 19. The s.c. has safe access to water                   |                  |               |                    |                     | 0.72         |          | 0.64        | 3.10 |
| Lighting                                                |                  |               |                    |                     |              |          |             |      |
| 20. The s.c. needs more lighting                        |                  |               |                    |                     |              | 0.72     | 0.55        | 3.40 |
| 21. I hope the s.c. has more users                      |                  |               |                    |                     |              |          | 0.32        | 3.90 |
| 22. The s.c. is a good place for children to play       |                  |               |                    |                     |              | 0.43     | 0.60        | 3.10 |
| 23. The water volume of the stream is enough            |                  |               |                    |                     |              | −0.46    | 0.59        | 3.00 |
| 24. I like the s.c.                                     | 0.61             |               | 0.43               |                     |              |          | 0.68        | 3.90 |
| 25. The stream area is a suitable place to live         | 0.43             |               | 0.43               |                     |              |          | 0.59        | 3.60 |
| Variance (%)                                            | 12.9             | 11.9          | 11.5               | 10.3                | 6.6          | 4.8      | 58.0        |      |

Factor loadings via varimax rotation >0.40 are listed; s.c.: stream corridor.



Table 4

Differences in standardized mean factor scores and various mean item scores by survey area

|                                            | Survey area |           |               |             |            | F-value | P-value |
|--------------------------------------------|-------------|-----------|---------------|-------------|------------|---------|---------|
|                                            | Yamahana R. | Shojin R. | Old Nakano R. | Yasuharu R. | Shinoro R. |         |         |
| Factor                                     |             |           |               |             |            |         |         |
| Recreational use                           | 0.07ab      | 0.08ab    | −0.09ab       | 0.25a       | −0.38b     | 3.57    | <0.01   |
| Participation                              | −0.26a      | 0.07a     | 0.18a         | −0.01a      | −0.13a     | 2.09    | 0.08    |
| Nature and scenery                         | 0.24ab      | 0.59a     | −0.14ab       | −0.48b      | 0.24ab     | 12.45   | <0.01   |
| Sanitary management                        | 0.30a       | 0.01ab    | −0.24b        | 0.15ab      | −0.05ab    | 3.33    | <0.01   |
| Water safety                               | 0.23a       | 0.17a     | −0.15a        | 0.47a       | −0.88b     | 18.70   | <0.01   |
| Item                                       |             |           |               |             |            |         |         |
| Connectivity of greenery (9 <sup>a</sup> ) | 3.57ab      | 3.98a     | 2.99b         | 2.98ab      | 3.06ab     | 9.95    | <0.01   |
| Good scenery (10)                          | 3.40a       | 3.58a     | 3.30a         | 3.09a       | 3.22a      | 1.88    | 0.11    |
| Good natural environment (11)              | 3.30a       | 3.44a     | 2.98ab        | 2.65b       | 3.37a      | 8.18    | <0.01   |
| No litter (13)                             | 2.87a       | 2.46ab    | 2.26b         | 2.72ab      | 2.41ab     | 3.31    | <0.05   |
| Well maintained (17)                       | 3.76a       | 3.60a     | 3.22ab        | 3.59a       | 2.78b      | 7.99    | <0.01   |
| Preference (24)                            | 4.02ab      | 4.28a     | 3.77b         | 3.91ab      | 3.53b      | 4.50    | <0.01   |

Standardized factor scores (mean = 0, S.D. = 1) were used for the five factors. Values followed by the same letters show no significant difference at the 5% level according to Scheffe's multiple comparison procedure.

<sup>a</sup> Corresponding number of the scale on the questionnaire (see Table 3).

### 3.3. Group differences

It can be easily appreciated that the public does not share a singular attitude toward stream corridors. Variance analysis using the standardized factor scores of a number of respondents' background variables showed that younger people's scores for participation were lower, with the scores increasing with age ( $F = 11.80$ ,  $P < 0.01$ ). Participation scores were also higher for residents who had lived in the area for a long time. While men on the whole had higher scores for this factor and sanitary maintenance than women ( $t = -4.44$ ,  $P < 0.01$  and  $t = -2.36$ ,  $P < 0.05$ , respectively), women had higher scores for nature and scenery ( $t = 2.05$ ,  $P < 0.05$ ).

Using cluster analysis of the factor scores, the respondents were classified into three groups of similar size as shown in Table 5. Group 1 had relatively low factor scores from recreational use to water safety, meaning that people from this group did not have a positive perception of the streams. Group 2 had high factor scores for recreational use and nature and scenery, while Group 3 was characterized by higher scores of participation and sanitary maintenance. For ease of recognition purposes, these groups are named the Passive Group, the Recreation Interested Group, and the Participation Oriented Group, respectively.

If respondents' backgrounds factors were to be taken into account, age may be seen as having had a great influence in this classification. Fifty people

Table 5

Differences in the factor scores according to the group

| Factor              | Group   |                       |                        | F-value | P-value |
|---------------------|---------|-----------------------|------------------------|---------|---------|
|                     | Passive | Recreation Interested | Participation Oriented |         |         |
| Recreational use    | −0.32b  | 0.58a                 | −0.17b                 | 28.30   | <0.01   |
| Participation       | −0.55c  | −0.03b                | 0.55a                  | 46.44   | <0.01   |
| Nature and scenery  | −0.47c  | 0.57a                 | −0.02b                 | 35.46   | <0.01   |
| Sanitary management | −0.43b  | −0.45b                | 0.78a                  | 86.83   | <0.01   |
| Water safety        | −0.33b  | 0.06a                 | 0.27a                  | 12.07   | <0.01   |

Standardized factor scores (mean = 0, S.D. = 1) were used for the five factors. Values followed by the same letters showed no significant difference at the 5% level according to Scheffe's multiple comparison procedure.

and over accounted for 40% of the Passive Group, 37% of the Recreation Interested Group, and 69% of the Participation Oriented Group, respectively. On the other hand, the Passive Group and the Recreation Interested Group included relatively young residents. A characteristic of the Participation Oriented Group was its high ratio of men to women (men: 55%, women: 45%) in contrast to the Passive Group and the Recreation Interested Group (men: 38 and 32%, respectively). The Participation Oriented Group may have included many retired people who have a lot of free time while the Recreation Interested Group, for its part, included a large number of people who use the stream corridors relatively often. Fifty-eight percent of respondents in this group used the stream corridors more than once per month, while the Passive and Participation Oriented Groups rated at 38 and 48%, respectively.

In terms of the distribution of the three groups over the surveyed areas, the Passive Group numbers were smallest at the Shojin (18%) and Yamahana Rivers (23%) while its ratio was highest at the Old Nakano River (43%). The ratio of the Recreation Interested Group was highest at the Shojin River (44%). At the same time, larger ratios for the Participation Oriented Group were found at the Yamahana and Yasuharu Rivers (43%, respectively). While it appears that residents' interest in these streams varied by location, it would be interesting to know the effects of vegetation on these perceptions.

### 3.4. Vegetation and perceptions

The outline of the vegetation survey results is shown in Table 6. At the low water channel terrace, water level, and in the surrounding environment, maintenance of vegetation may have had an influence on the number and variety of species found. However, the effect of water levels on the average number of species and ratio of exotic species per quadrat is not so clear; many species, especially native ones, were observed at a number of sites where the water level was high such as at the Old Nakano River. This may be due not only to water level, but also to a lack of mowing as well as the planting of native species when the river was renovated. Alternatively at the Yasuharu River, where the water level was relatively high, the ratio of exotic species was also found to be high. On the other hand, at the Shinoro River sites where water levels were low, many more native species including wetland ones were found, and the ratio of exotic species was low. Succession of vegetation from low herbaceous communities to tall herbaceous communities and shrubs was found at some sites such as at the Shinoro and Old Nakano Rivers.

At the high water channel terrace and bank slope, the mean number of species was from 4.5 to 5.9. The ratio of the exotic species differed greatly from site to site. For instance, the Yasuharu and Old Nakono Rivers, where mowing had been done once per year, had high ratios of 82 and 69%, respectively, whereas

Table 6

Occurrence of all species at the low water channel, high water terrace, and bank slope of the surveyed stream corridors with the breakdown of exotic to native species

| Survey area                       | Number of species per quadrat |                |                | Ratio of exotic species | Water level (cm) |
|-----------------------------------|-------------------------------|----------------|----------------|-------------------------|------------------|
|                                   | All species                   | Native species | Exotic species |                         |                  |
| Low water channel terrace         |                               |                |                |                         |                  |
| Yamahana River                    | 6.7                           | 4.3            | 2.4            | 36.1                    | −7.4             |
| Shojin River                      | 4.3                           | 3.3            | 1.0            | 23.5                    | −8.5             |
| Old Nakano River                  | 11.0                          | 10.7           | 0.3            | 3.0                     | 9.9              |
| Yasuharu River                    | 7.7                           | 3.7            | 4.0            | 52.2                    | 2.4              |
| Shinoro River                     | 20.5                          | 16.8           | 3.8            | 18.3                    | −20.5            |
| High water terrace and bank slope |                               |                |                |                         |                  |
| Shinoro River                     | 4.9                           | 3.8            | 1.1            | 21.6                    | −                |
| Old Nakano River                  | 5.9                           | 1.8            | 4.1            | 69.3                    | −                |
| Yasuharu River                    | 4.5                           | 0.8            | 3.7            | 82.2                    | −                |
| Shinoro River                     | 5.9                           | 4.2            | 1.7            | 28.8                    | −                |

There was no vegetation at high water terrace and bank slope of the Yamahana River. Water level was only surveyed at low water channel terrace.

the Shojin and Shinoro Rivers had relatively low ratios of 22–29%. High ratios of exotic naturalized species were found at sites where mowing had been done once or twice per year. Similar to the low water channel terrace, succession of vegetation from low herbaceous communities to tall herbaceous communities and shrubs was found at some sites where no mowing had been done such as at the Shinoro River and a part of the Shojin River.

Examination of these vegetation characteristics among the stream corridors reveals some corresponding relationships to the perceptions. The factor scores for Naturalness and Scenery at the Shojin and Shinoro Rivers, with low exotic species ratios, were higher than those of the Yasuharu River where mowing had lead to a greater frequency of exotic species. The Yasuharu River did, however, have a higher factor score for recreational use due to its having been opened by mowing. The factor scores for water safety were low at the Shinoro River where some tall native species such as reed grass and shrubs had occurred by succession. The low factor score for water safety at the Old Nakano River may have been due to the presence of taller native species at the low water channel.

#### 4. Discussion and application

Relatively high preference and high concern of residents toward the stream corridors in these greenways showed that the improvements made to the streams were appreciated by local residents. The results further suggested that any future stream corridor improvements would be substantially supported by nearby residents as shown by Tapsell (1995). This study has found the five factors of recreational use, participation, nature and scenery, sanitary management, and water safety as being important factors relating to public perception of urban stream corridors and greenways. Hashimoto and Ejiri (1993) reported that residents had three views of small rivers the “dwelling with respect to floods, etc.” view, the “visual” view, and the “ecological” view. If we apply these three factors to this study, water safety may be considered as being similar to “dwelling with respect to floods, etc.,” while the “visual” and “ecological” factors may be considered jointly as nature and scenery.

Of the five factors, the mean scores of the four factors aside from participation were significantly different among the survey areas due to the differing stream features. It may be suggested that the main purposes of the improvement works were accordingly different at the various streams, and that these characteristics of the stream corridors were reflected in the factors. For instance, the Yasuharu River, which had the highest scores for recreational use, was likely improved for easy accessibility to water including children’s recreation. Meanwhile the main purpose of improvements to the Shojin River, which had highest mean scores for nature and scenery, might have been environmental restoration. The relationships between the results and the greenways are further discussed below.

It was found that the main recreational activities in the stream corridors were strolling and relaxation. When stream corridors are renovated, usually a foot-path is built with some benches and shelters set up at various rest spots. However, for many of the stream corridors like the ones surveyed, only a part of the stream corridors have been improved and the connections between trails is poor in many cases. Although the trail systems are important in greenway planning, the trails in the Sapporo area rarely connect to other rivers or green spaces in an integrated manner. The five stream corridors in this study flow into larger rivers, but only the trail at the Shojin River connects to the Toyohira River side trail, which in turn is the main city-wide greenway. The Yamahana River trail which could potentially be made to link to the Toyohira River trail does not have a good connection because of barriers such as roads with heavy traffic. It is important to connect the trails and/or bikeways with the city-wide greenway. At the same time, it is necessary to make the trail and bikeway systems at the city-wide scales, which has not yet been well planned.

In this study, it was found that a sense of nature was not a separate issue from scenery (aesthetics). However in detail, the mean scores for “good natural environment” were significantly different among the survey areas but those for “good scenery” were not. In order to achieve highly preferred scenery in these stream corridors, there are three necessary components of natural scenery: water, vegetation, and sequential experience with variety. The high ratings for nature and scenery at the Shojin River can be interpreted as being rooted in these components. Behind the high

sense of nature at the Yamahana River, where there is no vegetation at the banks or high water channel terrace, is the presence of the nearby forest of Mt. Moiwa and other greenery. On the other hand, some stream corridors like the Yasuharu and Old Nakano Rivers had low “good natural environment” scores along with relatively high “good scenery” scores because of some good viewpoints with panoramic views of mountains located to the west. The mountains serve as important landmarks demarcating the edge of Sapporo. In densely built-up urban areas such as Sapporo, open and panoramic views appear to be preferred.

The preference scale had a high correlation with the “good natural environment” scale. This corresponds to previous studies which showed that people preferred natural scenery (e.g. Kaplan and Kaplan, 1989; Ulrich, 1986; House and Fordham, 1997). Although, the “well maintained” scale was included in sanitary maintenance (with the loading being not so high), it had relatively high correlation with the preference scale. Scores for “well maintained” were higher at the Yasuharu River and lower at the Shinoro River, a difference which may have been influenced by vegetation maintenance (mowing or lack thereof). Mowing may allow for easier water access, and recreational use as well as unobstructed views of the surrounding area. The relatively high preference rating of the Yasuharu River where mowing took place, and the low preference rating of the Shinoro River with its un-mowed, tall vegetation may be due to this. These results provide evidence for the idea that people prefer “controlled” or “refined” versions of nature, at least in urban contexts (Tapsell, 1995; Nassauer, 1997).

A natural river will constantly change creating banks, cliffs, and beaches while meandering across the landscape. Aside from the Shojin River, the stream corridors did not offer much in the way of diversity. This highlights the importance of considering not only the variety of shapes and dynamic changes which occur at streams, but also the effect maintenance may have on the diversity at target stream corridors. For vegetation diversity, and although it was not so clearly shown in Table 6, the water level and the flow of water are important elements at the low water channel. It was suggested that the presence of a number of plant species may increase the “feeling of nature” at stream corridors, while a high ratio of exotic species may decrease this feeling among residents. High ratios of

exotic species were found at some banks and high water channel terraces where mowing had taken place. Although the size and features of the stream areas are different, it may be argued that some areas should be managed so as to minimize human use and enhance natural richness (Manning, 1997). It may further be argued that to make clear connections between greenways, some areas without mowing should be planned to encourage riparian forest by natural succession. The Shinoro River, located at the urban fringe, has many native species which had taken root after renovation of the stream. The survey results showed that connectivity to surrounding natural areas is also important for promoting biodiversity. As such, greenway design should minimize barriers between the river valleys and adjacent vegetation patches (Baschak and Brown, 1994).

Meanwhile, as shown by the low water safety ratings at the Shinoro and Old Nakano Rivers, some spaces where people come in contact with water need maintenance for open viewing or easy access to water. For public safety, care should be taken to ensure that clear lines of sight are maintained along trails and around viewpoints (Luymes and Tamminga, 1995). To avoid conflicts between recreational use and biodiversity, separating spaces intended for recreation from those designated for natural preservation is one method that may be suggested. Manning (1997) pointed out that a “gradient” from extensive naturalness to intensive use is an important idea in both balancing out greenway planning and in vegetation management. The balance between naturalness and recreational use may differ according to the location of a given stream corridor. If the stream corridor were located in a central business district or near a symbolic landscape, then intensive recreational spaces would be more important. On the other hand, on the fringes of urban areas, natural conservation and the role of a biological corridor would be more important.

The results showed that residents were concerned about water quality and quantity. As mentioned before, many streams in the Sapporo which were fed by springs lost their water sources as many springs had dried up with urbanization. One current source of water is treated sewage water which occurs in the Yasuharu River. Other sources include water from other rivers, while some stream restoration projects use pumped water. The danger of flooding at these

streams is low, which was also reflected in the survey results of local residents. However, residents who are not aware of their nearby stream's risk of flooding or who live along streams where the water is not controlled may be more afraid of flooding. Thus, providing information about flood danger is important in formulating a stream corridor greenway plan.

At the same time, the effective use of rain water is also necessary. The stream corridor, including the area around the stream and buffer zone, serves as important protection for the stream. Although it is difficult to have enough space for the buffer zone in urban areas, trees on the street and the greenery of private sites along the stream contribute positively to the stream's environment. While as a general rule rain water should be channeled to the streams whenever possible, pollution control of surface run off water directed away from streets and urbanized areas is also necessary. In Sapporo, which has a lot of snow in the winter, the discarding of excess snow into the streams is also a serious problem. Hence it is necessary for retention basins and snow stock spaces, which act as filters, to be incorporated into the planning of stream corridor greenways, which are conserved as natural areas or used for recreational activities in other seasons.

In the survey results of this study, a large number of respondents gave positive responses with respect to the participation factor. Many people were anxious about litter, water pollution, and crime prevention. The most popular kinds of maintenance activities were cleaning the streams and mowing the grass, and while it may be easily suggested that a large part of residents' anxiousness would be relieved by promoting these maintenance activities, consideration of growing concerns toward streams and the general aquatic environment in a given area is also important.

From the results, three groups of respondents were classified and the ratios of each were found to differ among the survey areas due perhaps to differing features of the stream corridors and characteristics of the neighborhood communities (e.g. community activities, development types, and developed age of the areas). As for the kinds of participation programs suited to the three groups, some suggestions can be made corresponding with each group's unique characteristics. For the Passive Group who were not so interested in their nearby stream corridors, an environmental program which increases their interest in urban streams is

important. Also, the improvement of accessibility to the stream from their residences may be necessary. As for the Recreation Interested Group who visited and gave high ratings for nature and scenery for their surrounding stream corridors, their motivation to actively participate was relatively low. On average, members of the Interested Group were relatively younger than those of the Participation Oriented Group, and may have had less time to participate in stream corridor activities. It may be important to offer this group more opportunities to participate, including more family oriented activities in which age and sex would not be a barrier to participation. Concerning total water circulation, environmental programs aimed at reducing residents' water pollution and promoting water conservation are important for the Participation Oriented Group whose members have a high willingness to participate. These programs may form the basis of a strong background for promoting a stream corridor greenway plan. While the willingness of residents to cooperate with and support greenway planning may be seen, river improvement plans to this point have not been open enough to residents. In this way, it is necessary to offer local residents enough information and opportunity to participate in the stream corridor greenway planning process.

In Japan, there is no direct legal regulation to protect the buffer areas around stream corridors. Thus streams such as the Shojin River where its steeply sloped riparian forest is naturally preserved from urban development are very rare in urban areas. This study has shown that the effects of such buffer areas are well perceived by local residents. Scenic overlay zones ("scenic districts") set aside for the conservation of good scenery and greenery by the City Planning Law will contribute to the preservation and creation of greenery. Designated "green conservation sites" set out by the Conservation of Green Area Law, which may be applied to private land, will similarly contribute to the conservation of greenery. Although these laws may be useful for the planning of stream greenways, they will require residents' cooperation. The importance of land owners' cooperation to rural greenway planning was shown by Schrader (1995). In an urban stream corridor greenway plan, residents who own small housing sites along the stream may have varying attitudes toward cooperating with the plan. The three types of respondent groups outlined in this

study may provide a basis for further surveys in this area.

As pointed out by [Baschak and Brown \(1994\)](#), many improvements are made at isolated parts of a stream corridor. Furthermore, most stream improvements are typically done only at the stream terrace and banks, and these rarely coincide with planning for open spaces or land use along the streams. The main reason for this may be the sectionalism of involved administrations and the concentration of flood control improvements. The participation of residents in greenway planning would make it easier to take broader views, which assess the natural and cultural resources of a given area. When the greenway is being planned, an overall framework and the development of suitable assessment procedures for the recreational, ecological, and cultural resources of stream corridors are necessary as shown by [Fabos et al. \(1993\)](#).

## 5. Conclusion

Although many streams have disappeared or deteriorated in urban areas, a number of cities in Japan tend to integrate many stream corridors into greenway after improvements are made to them. This study has attempted to identify nearby residents' perceptions toward a number of improved stream corridors in Sapporo for greenway development. High preference and concern on the part of residents toward the stream corridors in question have shown that improvements to the stream corridors are supported by residents, and also that stream corridors have a great potential for greenway in the urban areas. There are, however, some problems regarding the implementation of a greenway plan owing to a lack of a comprehensive approach. The survey results have shown that the main factors accounting for residents' perceptions toward the stream corridors were recreational use, participation, nature and scenery, sanitary maintenance, and water safety. Aside from the participation factor, the mean scores of the four factors were significantly different among the stream corridors because of differences in the stream corridors' features and surrounding environmental conditions. Furthermore, the two factors of recreational use and nature and scenery were found to be highly related to the preferences of residents.

Based on the typology of greenways designed by [Ahern \(1995\)](#), the scale, landscape contexts, goals, and planning strategies of the stream corridors greenways in this study are as follows. The scale ranged from site-wide to city-wide, and the landscape context consisted of urban areas (mainly residential) and urban–rural fringe areas. Furthermore, it was found that the topography (alluvium fan and lowland), distance from hills and mountains, as well as surrounding land uses were also important contexts in the rating of nature and scenery. Although the main goals of the stream corridor greenway are recreation, ecology and water resources related, many improvements to the stream corridors have taken place at isolated parts of the stream corridors, and the goals of the improvements may have depended on the parts. Thus, a number of the stream corridors in this study where improvement goals were recreational use had lower ratings for nature and scenery, exemplified by the lower perceptions of nature evident at the Yasuharu River. Balancing recreational use and natural scenery is important for integrating stream corridors into a greenway system. However, more definitive study of the relationship between preference and vegetation is needed, it has been suggested that the number of plant species, exotic species ratio, and vegetation management have an influence on the perceptions of nature. In all, the goals of the stream corridor greenways should be more extensive and varied, and should be based on the characteristics of the streams, landscape contexts, and the perceptions of the residents.

When designing stream corridor greenways and connecting them to city-wide greenways, the following three points are important: (1) enhancing the ecological environment of the streams; (2) creating recreational circulation systems (trails, bikeways, and cross-country ski trails); and (3) making full use of the natural and cultural characteristics of the local areas. In turn, the following five points are important planning strategies: (1) making or enhancing the stream buffer; (2) the improvement of stream areas for ecological purposes including maintenance plans concerning biodiversity; (3) making or obtaining new green spaces or parks along the streams; (4) connecting the streams to other larger natural areas like hills, green areas, etc.; and (5) acquiring and maintaining water quantity. To realize these goals and strategies, public participation in greenway planning is essen-

tial. Based on the surveyed factor scores, respondents were classified into three groups; the Passive Group, the Recreation Interested Group, and the Participation Oriented Group. For promoting greenway development, these classifications of residents and appropriate involvement and management programs for them are essential for successful greenway planning along urban stream corridors.

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